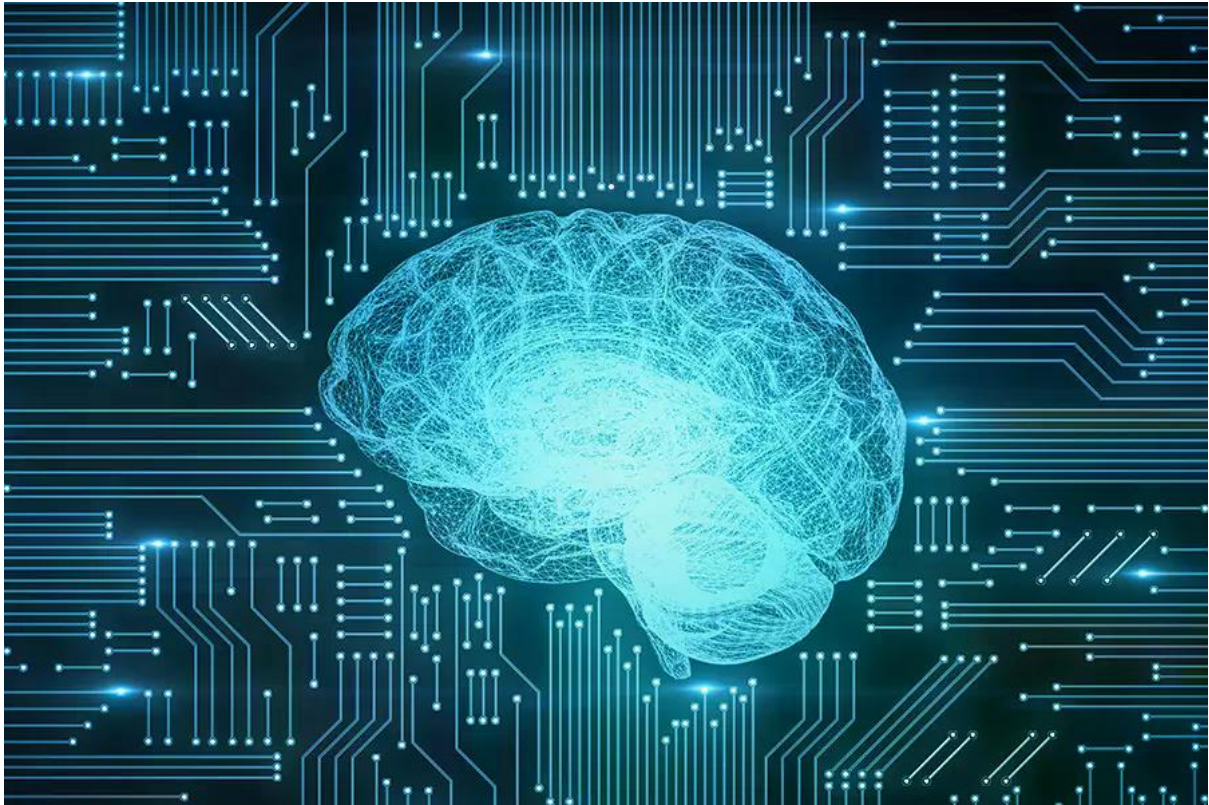




INTELLIGENT INFORMATION SYSTEMS 2026

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Chapter 1

Introduction

Information is a central concept of this book. The term is used in the title of the book, in this section, and in every chapter. To be an effective manager in any area of business, you need to understand that information is one of an organization's most valuable resources. Information is not the same thing as data, and knowledge is different from both data and information. These concepts will now be explained.

Data, Information, and Knowledge

Data consists of raw facts, such as an employee number, total hours worked in a week, an inventory part number, or the number of units produced on a production line. As shown in Table 1.1, several types of data can represent these facts. Information is a collection of data organized and processed so that it has additional value beyond the value of the individual facts. For example, a sales manager may want individual sales data summarized so it shows the total sales for the month.

Providing information to customers can also help companies increase revenues and profits. For example, social shopping Web site Kaboodle brings shoppers and sellers together electronically so they can share information and make recommendations while shopping online. The free exchange of information stimulates sales and helps ensure shoppers find better values.

TABLE 1.1
Types of data

Data	Represented By
Alphanumeric data	Numbers, letters, and other characters
Audio data	Sounds, noises, or tones
Image data	Graphic images and pictures
Video data	Moving images or pictures

Another way to appreciate the difference between data and information is to think of data as the individual items in a grocery list—crackers, bread, soup, cereal, coffee, dishwashing soap, and so on. The grocery list becomes much more valuable if the items in the list are arranged in order by the aisle in which they are found in the store—bread and cereal in aisle 1, crackers and soup in aisle 2, and so on. Data and information work the same way. Rules and relationships can be set up to organize data so it becomes useful, valuable information.

The value of the information created depends on the relationships defined among existing data. For instance, you could add specific identifiers to the items in the list to ensure that the shopper brings home the correct item—whole wheat bread and Kashi cereal in aisle 1, saltine crackers and chicken noodle soup in aisle 2, and so on. By doing so, you create a more useful grocery list.

Turning data into information is a process, or a set of logically related tasks performed to achieve a defined outcome. The process of defining relationships among data to create useful information requires knowledge, which is the awareness and understanding of a set of information and the ways in which that information can be made useful to support a specific task or reach a decision.

In other words, information is essentially data made more useful through the application of knowledge. For instance, there are many brands and varieties of most items on a typical grocery list. To shop effectively, the grocery shopper needs to have an understanding of the needs and desires of those being shopped for so that he knows to purchase one can of Campbell's (not the store brand!) low-sodium chicken noodle soup for the family member who is diabetic along with two cans of Campbell's regular chicken noodle soup for everyone else.

In some cases, people organize or process data mentally or manually. In other cases, they use a computer. This transformation process is shown in Figure 1.1.

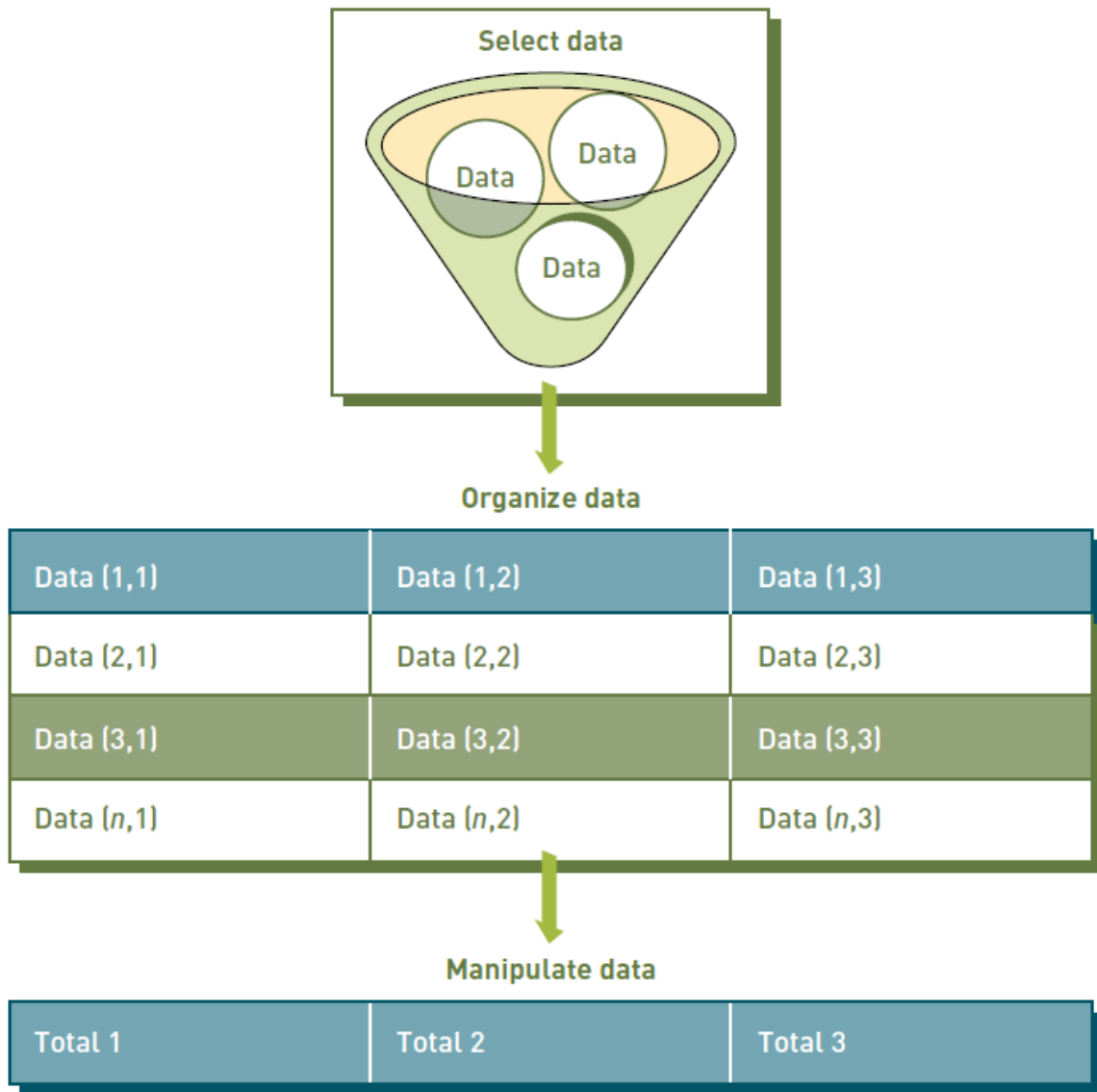


FIGURE 1.1
Process of transforming
data into information

The Value of Information

The value of information is directly linked to how it helps decision makers achieve their organization's goals. Valuable information can help people perform tasks more efficiently and effectively. Many businesses assume that reports are based on correct, quality information, but, unfortunately, that is not always true.

For example, Experian (a global information services firm that provides credit services, marketing services, decision analytics, and consumer services) estimates that on average, 22 percent of an organization's customer contact data is wrong. Companies can easily waste over \$100 per inaccurate customer contact data record on things like direct-mail marketing sent to wrong addresses and the inability to properly track leads. For an organization with 100,000 customers and 22 percent error rate, that projects to a loss of \$2.2 million.

Characteristics of Quality Information

Fundamental to the quality of a decision is the quality of the information used to reach that decision. Any organization that stresses the use of advanced information systems and sophisticated data analysis before information quality is doomed to make many wrong decisions.

Table 1.2 lists the characteristics that determine the quality of information. The importance of each of these characteristics varies depending on the situation and the kind of decision you are trying to make. For example, with market intelligence data, some inaccuracy and incompleteness is acceptable, but timeliness is essential. Market intelligence data may alert you that a competitor is about to make a major price cut.

The exact details and timing of the price cut may not be as important as being warned far enough in advance to plan how to react. On the other hand, accuracy and completeness are critical for data used in accounting for the management of company assets, such as cash, inventory, and equipment.

TABLE 1.2
Characteristics of quality information

Characteristic	Definition
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. This is commonly called garbage in, garbage out.
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the amount of money the company has invested in inventory.
Relevant	Relevant information is important to the decision maker. Information showing that lumber prices might drop is probably not relevant to a computer chip manufacturer.
Reliable	Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up may not be reliable.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

What Is an Information System?

Another central concept of this book is that of an information system. People and organizations use information systems every day. An information system (IS) is a set of interrelated components that collect, process, store, and disseminate data and information; an information system provides a feedback mechanism to monitor and control its operation to make sure it continues to meet its goals and objectives.

The feedback mechanism is critical to helping organizations achieve their goals, such as increasing profits or improving customer service. A computer-based information system (CBIS) is a single set of hardware, software, databases, networks, people, and procedures that are configured to collect, manipulate, store, and process data into information.

Increasingly, companies are incorporating computer-based information systems into their products and services. Investment companies offer their customers a wide range of powerful investment tools, including access to extensive online research. Automobiles are available with advanced navigation systems that not only guide you to your destination but also incorporate information regarding the latest weather and traffic conditions to help you avoid congestion and traffic delays. Watches, digital cameras, mobile phones, music players, and other devices rely on CBIS to bring their users the latest and greatest features.

The components of a CBIS are illustrated in **Figure 1.2**. An organization's technology infrastructure includes all the hardware, software, databases, networks, people, and procedures that are configured to collect, manipulate, store, and process data into information. The technology infrastructure is a set of shared IS resources that form the foundation of each computer-based information system.

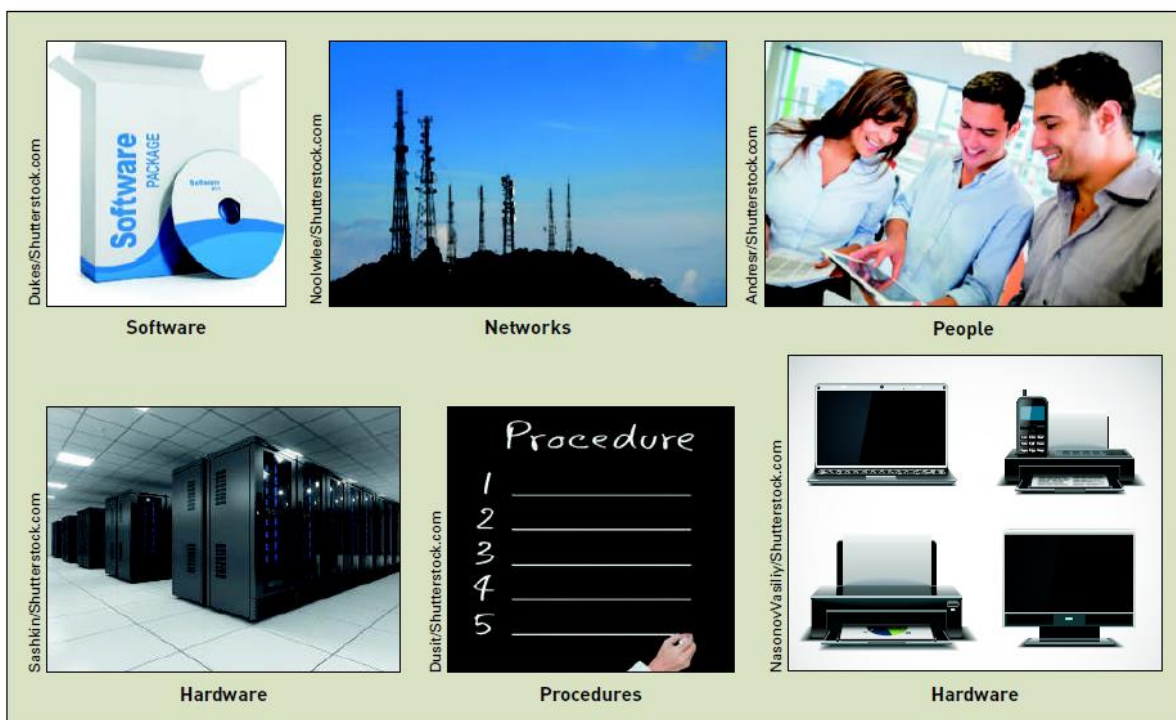


FIGURE 1.2
Components of a computer-based
information system

People make the difference between success and failure in all organizations. **Those who build great companies understand that the ultimate throttle on growth for any great company is not markets, or technology, or competition, or products. It is one thing above all others: the ability to get and keep enough of the right people.** Thus, it comes as no surprise that people are the most important element in computer-based information systems.

Good systems can enable people to produce exceptional results. They can also boost job satisfaction and worker productivity. Information systems personnel include all the people who manage, run, program, and maintain the system, including the chief information officer (CIO), who leads the IS organization. End users are people who work directly with information systems to get results. They include financial executives, marketing representatives, and manufacturing line operators.

A **procedure** defines the steps to follow to achieve a specific end result, such as enter a customer order, pay a supplier invoice, or request a current inventory report. Good procedures describe how to achieve the desired end result, who does what and when, and what to do in the event something goes wrong. When people are well trained and follow effective procedures, they can get work done faster, cut costs, make better use of resources, and more easily adapt to change. When procedures are well documented, they can greatly reduce training costs and shorten the learning curve.

Using a computer-based information system (**CBIS**) involves setting and following many procedures, including those for the operation, maintenance, and security of the system. For example, some procedures describe how to gain access to the system through the use of some log-on procedure and a password. Others describe who can access facts in the database or what to do if a disaster, such as a fire, earthquake, or hurricane, renders the CBIS unusable.

Good procedures can help companies take advantage of new opportunities and avoid lengthy business disruptions in the event of natural disasters. Poorly developed and inadequately implemented procedures, however, can cause people to waste their time on useless rules or result in inadequate responses to disasters.

Information Systems in Organizations

Most organizations have a number of different information systems. When considering the role of business managers in working with IS, it is useful to divide information systems into three types: personal IS, group IS, and enterprise IS.

Personal IS includes information systems that improve the productivity of individual users in performing stand-alone tasks. Examples include personal productivity software, such as word-processing, presentation, and spreadsheet software.

In today's fast-moving, global work environment, success depends on our ability to communicate and collaborate with others, including colleagues, clients, and customers. **Group IS** includes information systems that improve communications and support collaboration

among members of a workgroup. Examples include Web conferencing software, wikis, and electronic corporate directories.

Enterprise IS includes information systems that organizations use to define structured interactions among their own employees and/or with external customers, suppliers, government agencies, and other business partners. Successful implementation of these systems often requires the radical redesign of fundamental work processes and the automation of new processes. Target processes may include purely internal activities within the organization (such as payroll) or those that support activities with external customers and suppliers (order processing and purchasing). Three examples of enterprise IT are transaction processing, enterprise, and interorganizational systems.

For each type of IS, certain key **organizational complements** must be in place to ensure successful implementation and use of the system. These complements include:

- **Well-trained workers.** Employees must be well trained and understand the need for the new system, what their role is in using or operating the system, and how to get the results they need from the system.
- **System support.** Trained and experienced users who can show others how to gain value from the system and overcome start-up problems.
- **Better teamwork.** Employees must understand and be motivated to work together to achieve the anticipated benefits of the system.
- **Redesigned processes.** New systems often require radical redesign of existing work processes as well as the automation of new processes.

- **New decision rights.** Employees must understand and accept their new roles and responsibilities including who is responsible for making what decisions. Roles and responsibilities often change with introduction of a new system.

Managers have an essential role to play in the successful implementation and use of information systems. That role changes depending on which type of IS system is being implemented, as shown in **Table 1.3**, which also highlights other characteristics and provides examples of each type.

TABLE 1.3
Examples and characteristics
of each type of information system

	Personal IS	Group IS	Enterprise IS
Examples	Personal productivity software, decision-support system	Email, instant messaging, project management software	Transaction processing systems, enterprise systems, interorganizational systems
Benefits	Improved productivity	Increased collaboration	Increased standardization and ability to monitor work
Organizational complements (including well-trained workers, better teamwork, redesigned processes, and new decision rights)	- Does not bring complements with it - Partial benefits can be achieved without all complements being in place	- At least some complements must be in place when IS "goes live" - Allows users to implement and modify complements over time	Full complements must be in place when IS "goes live"
Manager's role	- Ensure that employees understand and connect to the change - Encourage use - Challenge workers to find new uses	- Demonstrate how technology can be used - Set norms for participation	- Identify and put into place the full set of organizational complements prior to adoption - Intervene forcefully and continually to ensure adoption

Chapter 2

Information Technology

Concepts

Hardware and Mobile Devices

Hardware consists of computer equipment used to perform input, processing, storage, and output activities. The trend in the computer industry is to produce smaller, faster, and more mobile hardware, such as smartphones, laptops, and tablet computers. In addition, hardware manufacturers and entrepreneurs are hard at work developing innovative new hardware devices, such as the following:

- Advanced keyboards that turn individual keys on a keyboard into trackpad-covered buttons where certain dual-purpose keys could be depressed to multiple levels to complete different tasks; the spacebar, for example, can serve its usual purpose. But add capacitive touch and it becomes a cursor; press a little harder to generate a mouse click. (Capacitive touch relies on the electrical properties of the human body to detect when and where on a display the user touches. Because of this, capacitive displays can be controlled with very light touches of a finger.)
- Laptops and displays that connect wirelessly, thus eliminating the need
- for expensive HDMI or DisplayPort display cables
- Computing devices with embedded 3D cameras, which will be able to recognize objects and even measure distances between things
- Keyboards that enable users to log in to Web sites via fingerprint authentication so they won't have to remember dozens of passwords for different sites
- Very-high resolution display devices that will show content in incredible detail and dramatically improve the viewing experience (think clarity and resolution way beyond 1080p HD)

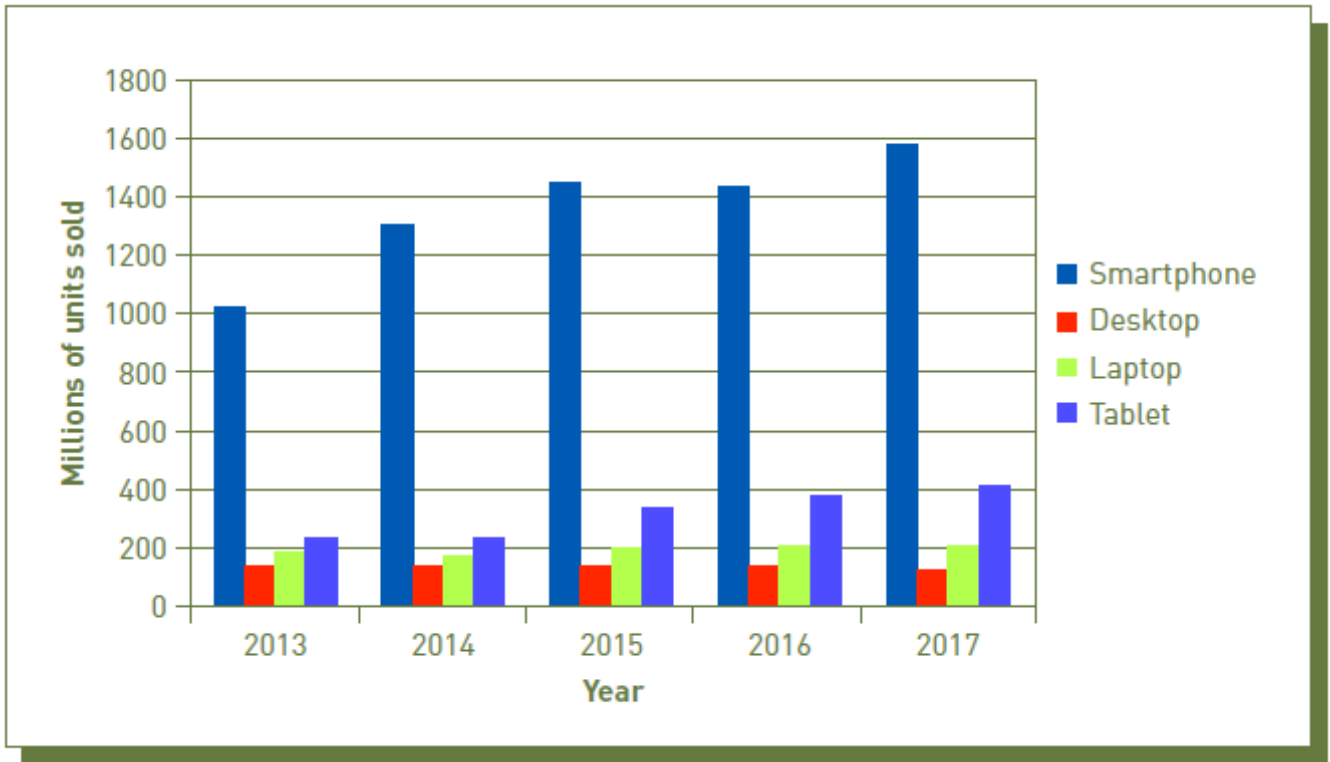
- **Computerized event data recorders (EDRs) that, like an airplane's black box, record vehicle speed, possible engine problems, driver performance, and more**

While desktop, laptop, and tablet computers continue to be used in a variety of settings, smartphones have become the primary device used by people around the world to communicate, go online, and access and share information.

In 2013, the number of smartphone users first exceeded the number of personal computer users, and the gap keeps growing, with the number of smartphones sold worldwide far exceeding the combined sale of desktops, laptops, and tablets as shown in **Figure 1.3. This rapid growth has been spurred by the improving affordability and capability of smartphones, the increasing speed and coverage of wireless networks, longer battery life, and the availability of hundreds of thousands of smartphone applications and games.**

For many people in developing countries, a smartphone is their first computer and their only Internet-connected device. For those in developed countries, it is common for individuals who do have a computer to also have a smartphone. It is projected that roughly one-third of the world's population will own a smartphone by 2018.

FIGURE 1.3
Millions of computing
Devices sold worldwide



Software and Mobile Applications

Software consists of the computer programs that govern the operation of a particular computing device, be it desktop computer, laptop, tablet, smartphone, or some other device. There are two types of software: system software and application software. System software—such as Google’s Android or Apple’s iOS—oversees basic computer operations such as start-up, controls access to system resources, and manages memory and files. Application software, such as Microsoft Office, allows you to accomplish specific tasks, including editing text documents, creating graphs, and playing games. Both system software and application software are needed for all types of computers, from small handheld devices to large supercomputers. In choosing application software, you must choose software that will work with the operating system installed on your computing device.

As of June 2015, 1.6 million applications were available for devices that run under the Android operating system and roughly the same (1.5 million) available for download from Apple’s App Store.^{8,9} The number of apps for each operating system is increasing by roughly 25,000 to 50,000 per month.

Business application software can be categorized by whether it is intended to be used by an individual, a small business, or a large multinational enterprise. For example, Quicken has long been a favorite accounting application for individuals who need money management and budgeting tools to help them watch their spending, increase their savings, and avoid late fees with alerts on upcoming payment due dates.

QuickBooks, an accounting application popular with small businesses, enables users to create invoices, track sales and expenses, process credit card payments, run payroll, and generate financial, tax, and sales reports. SAP ERP Financials is an accounting application used by many large, multinational organizations to manage the complexities of global accounting and reporting requirements.

SAP's software records all financial transactions in a comprehensive general ledger; supports sophisticated reporting requirements; provides management accounting tools for orders, projects, cost centers, and profit centers; enables the speedy and accurate closing of the firm's books; and helps manage risk and compliance across accounting and finance.

An important trend in the design of business application software is the attempt to imitate the look, feel, and intuitive ease of use associated with consumer apps that can be downloaded from the Google Play Store and the Apple App Store. Indeed, usability and user-interface design are key factors in delivering apps that business users will actually use.

In some organizations, employees can log on to enterprise app stores to acquire the latest company software and software upgrades. This trend of consumer technology practices influencing the way business software is designed and delivered is called the **consumerization of IT.**

CDW is a leading provider of integrated information solutions to corporate customers in small, medium, and large private and public organizations in the United States and Canada. The firm has established its App Marketplace Web site (<http://appmarketplace.cdw.com>) where customers' employees can identify leading enterprise mobile app solutions. The Web site is easy to

navigate with apps organized by industry and app functions. The apps there have already been vetted by CDW to perform effectively and reliably. In addition, for those needing custom mobile apps, the Web site identifies mobile app partners who have a proven track record of building, deploying, and updating top enterprise applications.

Database Systems and Big Data

A database is an organized collection of facts and information, typically consisting of two or more related data files. An organization's database can contain facts and information on customers, employees, inventory, sales, online purchases, and much more. A database is essential to the operation of a computer-based information system.

As anyone who works in marketing or sales knows, one of the biggest challenges that any business faces is the ability to generate new leads in an effort to locate customers. As businesses have looked for ways to meet this challenge, numerous vendors have seized the opportunity by offering access to databases of potential clients in various industries, as shown in [Table 1.4](#).

TABLE 1.4
Sample of marketing databases used
to generate sales leads

Industry	Database Vendor	Number of Records (Thousands)
Auto dealers	Oddity Software	110
Barber shops	Usable Databases	53
Dry cleaning/laundry	Oddity Software	42
Gas stations with convenience stores	CHD	56
Healthcare providers and decision makers	SK & A	2,100
Music retail outlets	Almighty Music Marketing	10
Nursing and retirement homes	MCH	34
Pet shops and pet supply stores	Oddity Software	15

A **data warehouse** is a database that stores large amounts of historical data in a form that readily supports analysis and management decision making. In a process called the **extract-transform-load (ETL)** process, raw data is extracted from various sources, transformed into a format that will support the analysis to be performed, and then loaded into the data warehouse. Data warehouses frequently hold a huge amount of data; they often contain five years or more of data. Many organizations employ data warehouses to hold the data they need to make key business decisions:

- Walmart operates separate data warehouses for Walmart and Sam's Club. Through these data warehouses, the company allows suppliers access to almost any data they could possibly need to determine which of their products are selling, how fast, and even whether they should redesign their packaging to fit more product on store shelves.

- Harrah's (part of the Caesar's Entertainment casino empire) uses a data warehouse to determine how much money particular gamblers are willing to lose in a day before they will decide not to come back the next day.
- Continental Airlines uses a data warehouse to help it determine who its most valuable customers are and to find ways to keep them satisfied—for example, by proactively making alternative travel arrangements for them if their flights get delayed.
- Macy's uses a terabyte-sized data warehouse to target improvements in four key areas of its e-commerce business: measuring the profitability and effectiveness of banner advertising, analyzing customer interactions and paths through its Web site, improving fulfillment capabilities, and correlating online sales with store sales to cross-sell and upsell customers across its distribution channels.

The digital universe (the collection of all data that exists) is doubling in size every two years as shown in **Figure 1.4**. Organizations are challenged by this rapid growth and at the same time scrambling to take advantage of the opportunities provided by this data. Big data is a term used to describe data collections that are so enormous (think petabytes or larger) and complex (from sensor data to social media data) that traditional data management software, hardware, and analysis processes are incapable of dealing with them.

To gain a perspective on the quantity of data some organizations are struggling to manage, consider that the amount of data traveling over mobile networks alone is expected to exceed 10 exabytes per month by 2016. Table 1.5 defines the units of measure for data.

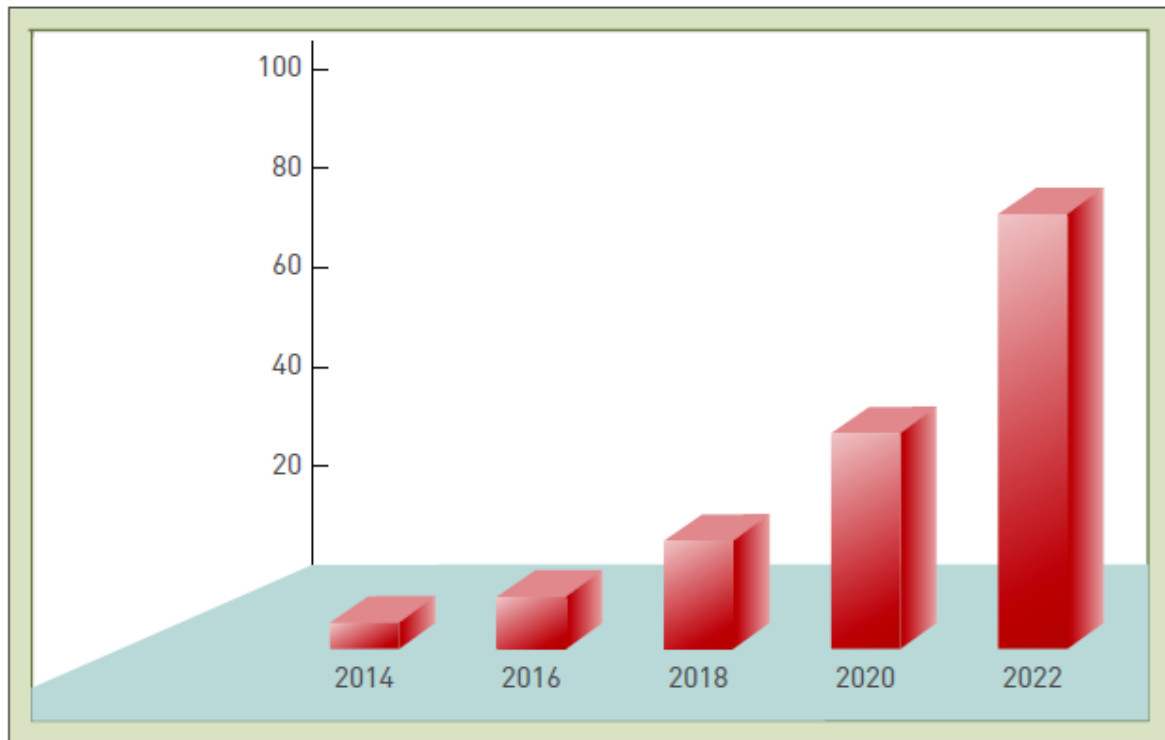


FIGURE 1.4

**The size of the digital universe
(zettabytes) doubles every two years**

TABLE 1.5

Units of measure for data

Unit of Measure	Size	Equivalent To
Byte	1 byte	One alphanumeric character
Kilobyte	1,000 bytes	The text of a joke or very short story
Megabyte	1,000 kilobytes	800 pages of text
Gigabyte	1,000 megabytes	7 minutes of HD-TV
Terabyte	1,000 gigabytes	The Hubble Space Telescope collected more than 45 terabytes of data in its first 20 years of observations
Petabyte	1,000 terabytes	50 years' worth of DVD-quality video
Exabyte	1,000 petabytes	44 billion 25 gigabytes Blu-ray discs
Zettabyte	1,000 exabytes	The amount of text created by every man, woman, and child on earth tweeting continuously for 100 years
Yottabyte	1,000 zettabytes	One thousand times the grains of sand on all of Earth's beaches

To avoid being paralyzed by information overload, organizations and indeed society itself must find a way to deal with this oncoming tsunami of data. This challenge has several aspects, including how to choose which subset of data to keep, where and how to store the data, how to find the nuggets of useful data that are relevant to the decision making at hand, and how to derive real value from the relevant data.

Ideally, the many challenges associated with big data will be conquered, and more data will lead to more accurate, insightful analyses and better decision making, which in turn, will result in deliberate actions and improved business results. For that to happen, society will need to address the many issues associated with big data, including concerns over invasions of privacy and the potential for overly intrusive monitoring of individuals by governments and organizations.

Networks and Cloud Computing

Networks connect computers and equipment in a room, building, campus, city, across the country, or around the world to enable electronic communication. Wireless transmission networks enable the use of mobile devices, such as smartphones and tablets. Telecommunication companies are now working on fifth-generation wireless communications that will enable transmission speeds 10 times faster than currently available on wireless networks—with greater coverage area and lower battery consumption—possibly as soon as the year 2020. Such technology will be needed to support the increased demand for faster transfer of data and video.

The Internet is the world's largest computer network, consisting of thousands of interconnected networks, all freely exchanging information. People use the Internet to research information, buy and sell products and services, email and instant message one another, participate in social networks (e.g., Facebook and LinkedIn), make travel arrangements, complete banking transactions, make investments, download music and videos, read books, and watch movies among other activities.

With **public cloud computing**, a service provider organization owns and manages the hardware, software, networking, and storage devices, with cloud user organizations (called tenants) accessing slices of shared resources via the Internet. The service provider can deliver increasing amounts of computing, network, and storage capacity on demand and without requiring any capital investment on the part of the cloud users. Thus, public cloud computing is a great solution for organizations whose computing needs vary greatly depending on changes in demand. Amazon, Cisco Systems, IBM, Microsoft, Rackspace, Verizon Communications Inc., and VMWare are among the largest cloud computing service providers. These firms typically offer a monthly or annual subscription service model; they may also provide training, support, and data integration services.¹⁶ Online content provider Netflix uses the Amazon Web Services (AWS) cloud computing service to provide global delivery of some 10 billion hours of content per month. AWS enables Netflix users anywhere in the world to stream TV shows and movies to computers and mobile devices.

The **World Wide Web (WWW)**, better known simply as “the Web,” is a network of links on the Internet to files containing text, graphics, video, and sound. Information about the documents and access to them are controlled and provided by tens of thousands of specialized computers called Web servers. The Web is one of many services available over the Internet, and it provides access to millions of files.

New Internet technologies and increased Internet communications and collaboration are collectively called Web 2.0.

The technology used to create the Internet is also being applied within organizations to create **intranets**, which enable communication, collaboration, search functions, and information sharing between the members of an organization's team using a Web browser. For example, the Swiss Medical Group, based in Buenos Aires, Argentina, is a network of professionals, clinics, and laboratories that provide healthcare services and postgraduate medical education programs. It employs an intranet to provide its 9,000 employees with communication tools, company and industry news from various sources, announcements regarding new assignments for employees, tutorials, an online phone directory, videos, and document management services.

An **extranet** is a network based on Web technologies that allows selected outsiders, such as business partners and customers, to access authorized resources of a company's intranet. Many people use extranets every day without realizing it—to order products from their suppliers, track shipped goods, or access customer assistance from other companies. Federal Express (FedEx) was one of the first large companies to empower customers to serve themselves at their convenience through the use of a corporate extranet. A fundamental FedEx belief is that the information it provides customers about its services is more important than the services themselves. Customers can access the FedEx extranet to obtain a full range of shipping, billing, and tracking services. See **Figure 1.5**.

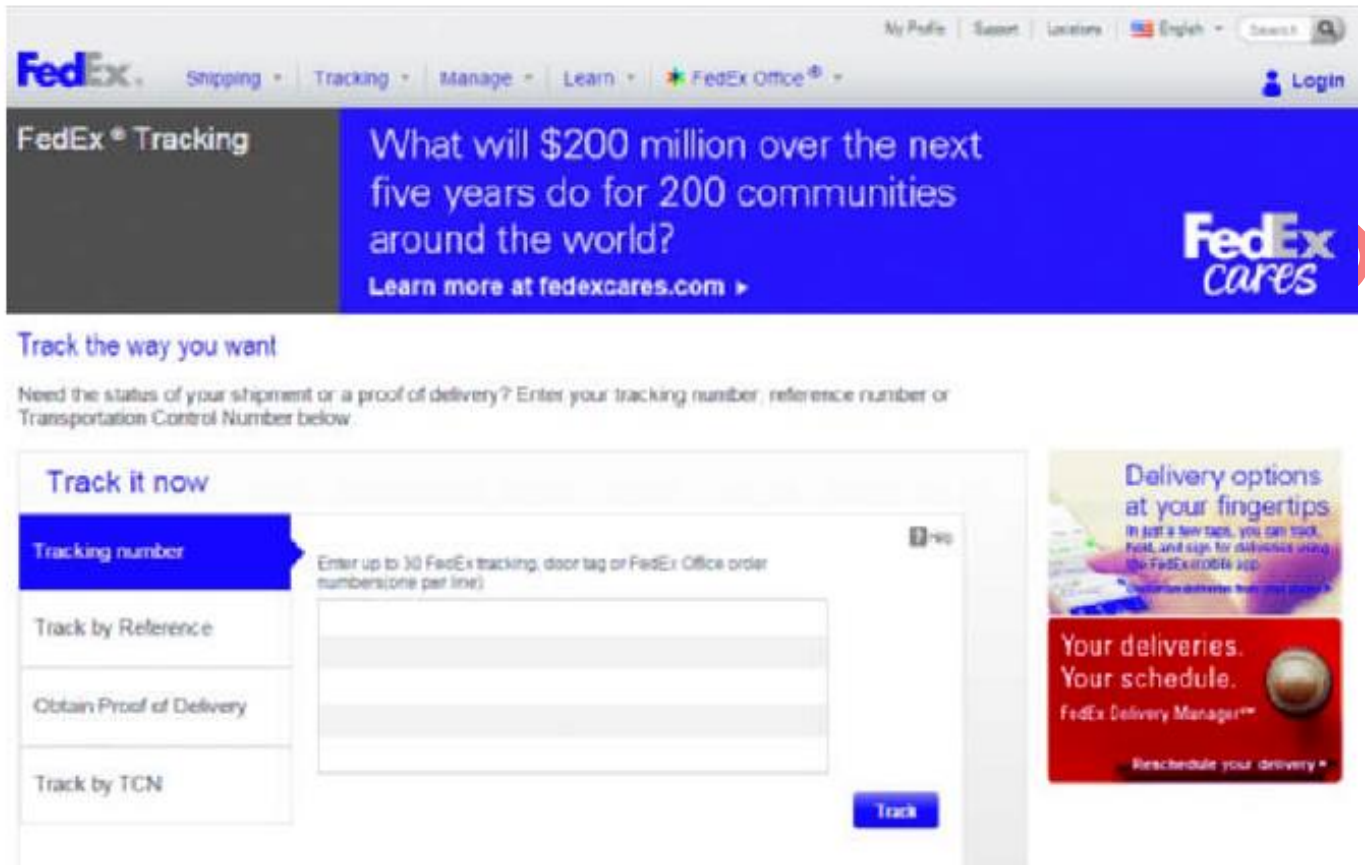


FIGURE 1.5

Extranets

The **Internet of Things (IoT)** is a network of physical objects or “things” embedded with sensors, processors, software, and network connectivity capability to enable them to exchange data with the manufacturer of the device, device operators, and other connected devices. In theory, the IoT would enable us to connect almost any device with an on/off switch to a network—automobiles, appliances, components of an aircraft engine, heart monitor implants, packing labels, ingestible pills, wearable devices, and even highway sensors that can warn of traffic and hazardous road conditions.

Each thing is uniquely identifiable and capable of interoperating with other “things” within the existing IoT infrastructure, often by connecting to a central hub. The IoT also includes cloud services, which enable the collection and analysis of data so people can process the data and take appropriate action via mobile apps.

Until recently, the IoT has been most closely associated with machine-to-machine communications, such as that employed in the manufacturing, gas, oil, and power industries. For example, in oil and gas drilling operations, remote sensors can measure important parameters such as pressure, flow rates, temperatures, and fuel levels in on-site equipment. These variables are transmitted to a computer that automatically adjusts the operation of the equipment to optimize hydrocarbon production, improve operational safety, and protect the environment.

The **Internet of Everything (IoE)** encompasses not only machine-to-machine but also people-to-people and people-to-machine connections. It is estimated that the total number of devices supported by the IoE could reach 50 billion by the end of 2020 See **Figure 1.6**. This rapid growth is being fueled by the increasing availability of network access, the creation of more inexpensive smart devices with sensors and network capabilities built into them, the rapid growth in smartphone penetration, and the creativity and innovation of people who are able to see and capitalize on the almost unlimited opportunities.

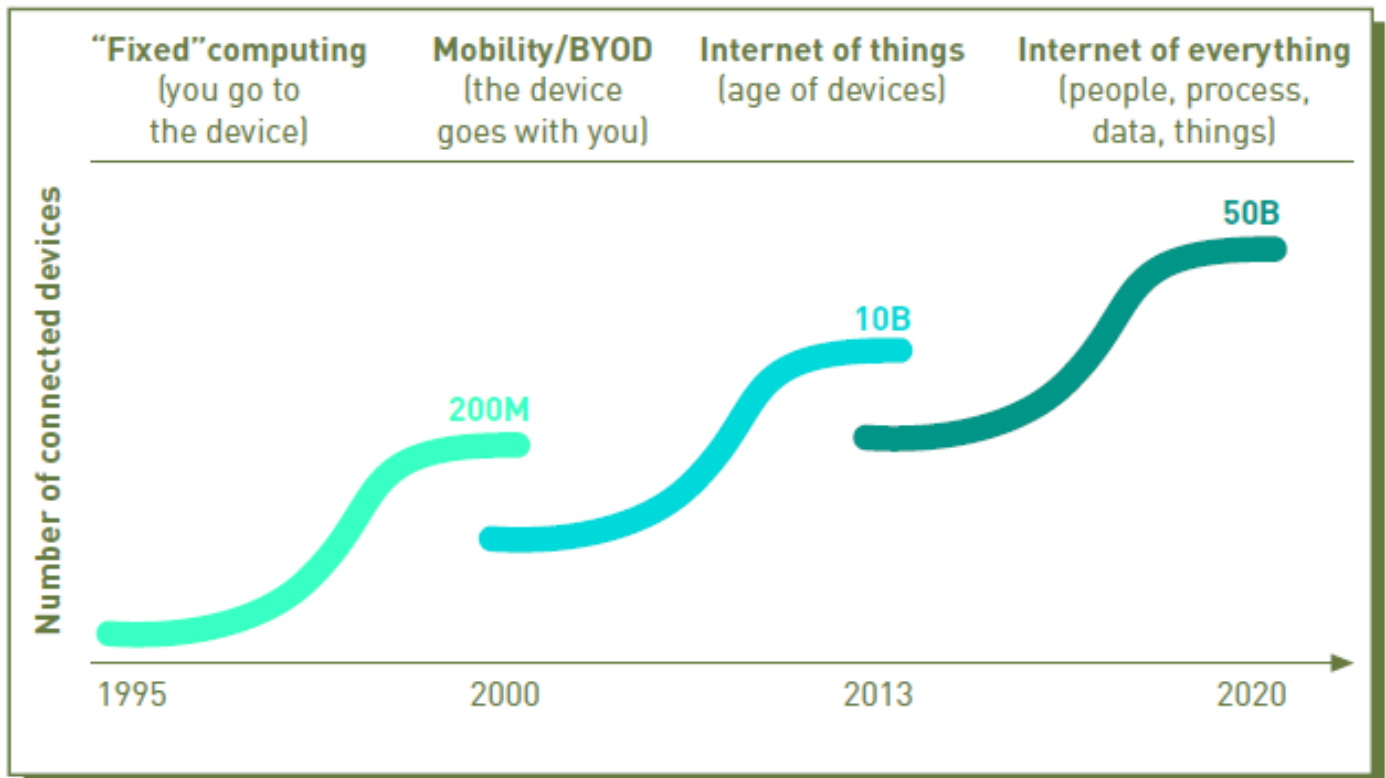


FIGURE 1.6
Growth of the Internet of
Everything (IoE)

General Electric (GE) is making a major strategic investment in the Internet of Things by offering its Predix Cloud service for industrial data and analytics. Predix will enable GE customers to connect sensors placed on industrial machinery such as jet engines, wind turbines, and locomotives to remote computing centers “in the cloud,” where the data from the sensors can be processed and analyzed to determine settings for optimal operating efficiencies and to schedule preventative maintenance to avoid unexpected failures.

Chapter 1

Information Systems

In Organizations

An **organization** is a group of people that is structured and managed to meet its mission or set of group goals. “Structured” means that there are defined relationships between members of the organization and their various activities, and that processes are defined that assign roles, responsibilities, and authority to complete the various activities. In many cases, the processes are automated using well-defined information systems. Organizations are considered to be open systems, meaning that they affect and are affected by their surrounding environment. See **Figure 2.1**.

Providing value to a stakeholder—customer, supplier, partner, shareholder, or employee—is the primary goal of any organization. The value chain, first described by Michael Porter in a classic 1985 Harvard Business Review article titled “How Information Gives You Competitive Advantage,” reveals how organizations can add value to their products and services. The **value chain** is a series (chain) of activities that an organization performs to transform inputs into outputs in such a way that the value of the input is increased.

An organization may have many value chains, and different organizations in different industries will have different value chains. As an example of a simple value chain, the gift-wrapping department of an upscale retail store takes packages from customers, covers them with appropriate, decorative wrapping paper, and gives the package back to the customer, thus increasing the customer’s (and the recipient’s) perceived value of the gift.

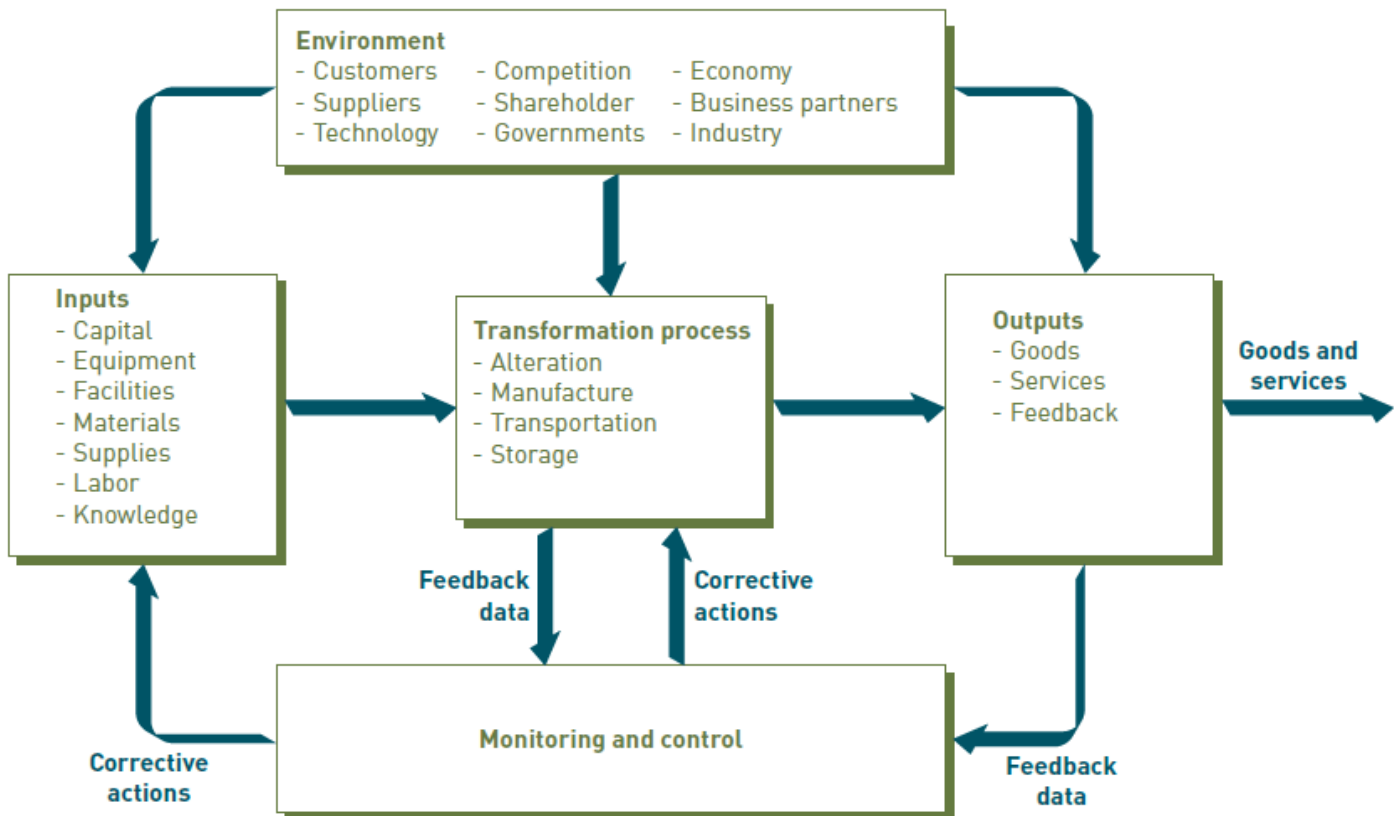


FIGURE 2.1
General model of an organization

In a manufacturing organization, the **supply chain** is a key value chain whose primary activities include inbound logistics, operations, outbound logistics, marketing and sales, and service. See **Figure 2.2**. These primary activities are directly concerned with the creation and/or delivery of the product or service. The supply chain also includes four main areas of support activities, including technology infrastructure, human resource management, accounting and finance, and procurement. (Technology infrastructure includes not only research and development but also information systems hardware, software, databases, and networks.)

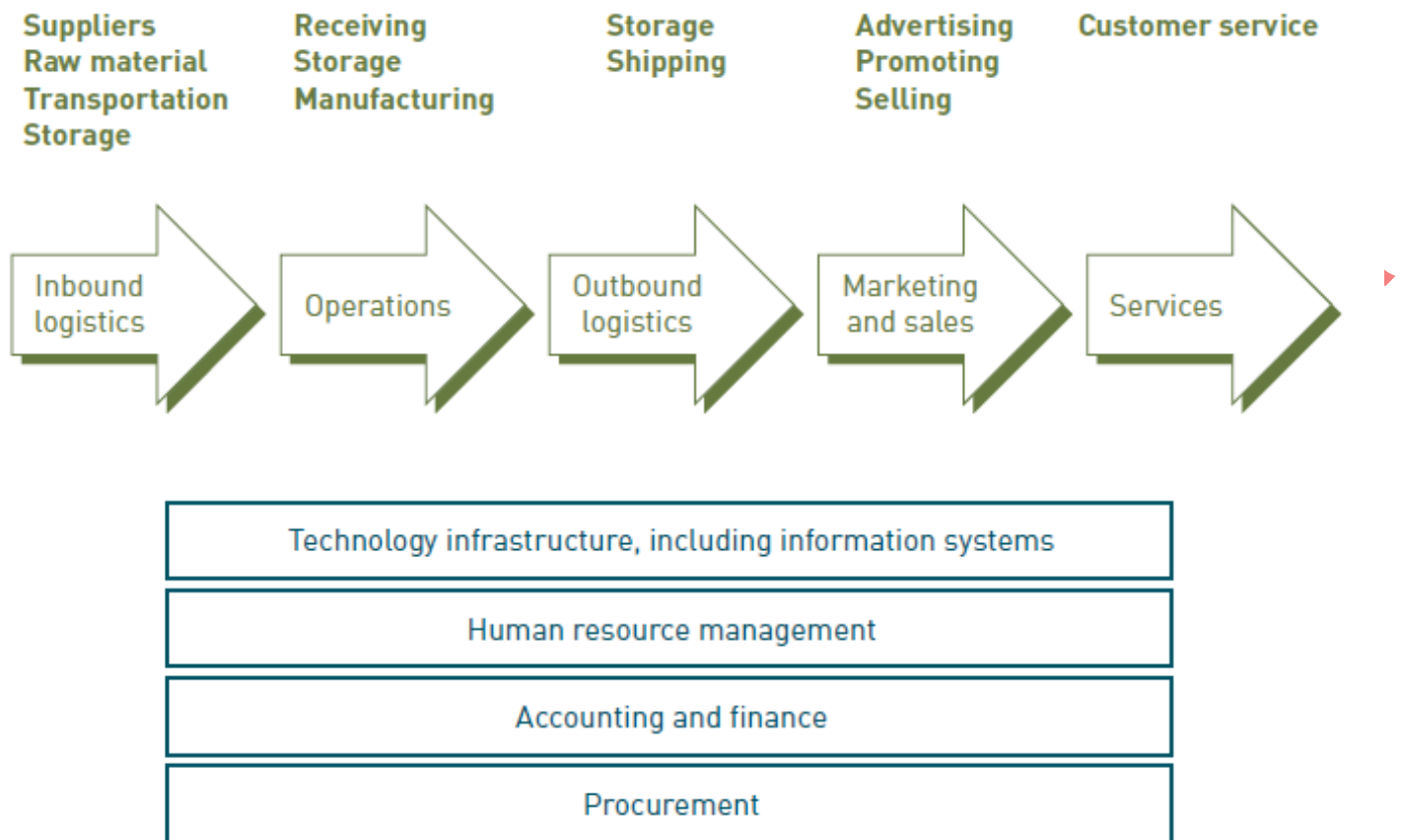


FIGURE 2.2

Supply chain

The concept of value chain is also meaningful to companies that don't manufacture products, including tax preparers, restaurants, book publishers, legal firms, and other service providers. By adding a significant amount of value to their products and services, companies ensure their success.

Supply chain management (SCM) encompasses all the activities required to get the right product into the right consumer's hands in the right quantity at the right time and at the right cost—from the identification of suppliers and the acquisition of raw materials through manufacture and customer delivery. The organizations that compose the supply chain are "linked" together through both physical flows and

information flows. Physical flows involve the transformation, movement, and storage of supplies and raw materials. Information flows allow participants in the supply chain to communicate their plans, coordinate their work, and manage the efficient flow of goods and material up and down the supply chain. See **Figure 2.3**.



FIGURE 2.3
Ford Motor Company
assembly line

Organizations are constantly fine-tuning and adjusting their supply chain. For example, many companies are increasing their use of free shipping to customers in hopes of increasing sales and profits. Amazon is experimenting with Amazon Fresh, a Web site that offers fast, free delivery of groceries on orders over \$35 and other products in the Los Angeles and Seattle areas. Customers can place an order by 10 am and have it by dinner; orders placed by 10 pm will be delivered by breakfast time. The e-commerce giant is also experimenting with a new drive-up store concept that will allow consumers to order grocery items online and then schedule a pickup at a dedicated facility.

Many organizations are also outsourcing much of their outbound distribution activities, including the storage and shipping of finished products to customers and the return of items from customers. Amazon, DHL, FedEx, Rakuten, Shipwire, UPS, and other companies are highly skilled and efficient at performing these functions.

What role do information systems play in supply chain management activities and other organizational activities? A traditional view of information systems holds that organizations use them to control and monitor processes and to ensure effectiveness and efficiency. In this view, information systems are external to the supply chain management process and serve to monitor or control it.

A more contemporary view, however, holds that information systems are often so intimately involved that they are part of the process itself. From this perspective, the information system plays an integral role in the process, whether providing input, aiding product transformation, or producing output. Zara and Coles are two examples of organizations that have incorporated information systems into the supply chain and made them integral parts of this process.

Zara is a Spanish clothing and accessories retailer with headquarters in Arteixo, Spain, and 2,000 stores spread across 88 countries.¹ Its founder, Amancio Ortega, had humble origins, but today is the third richest man in the world. Consumer clothing trends are constantly changing, creating a highly competitive environment in which companies compete not only on price but also on their ability to deliver products that are new and stimulating to their customers. To meet this challenge, Zara has developed an extremely responsive supply chain that enables it to go from design stage to sales floor in a maximum of three weeks rather than the six-month industry average. Zara can deliver new products twice a week to its stores around the world.

Mobile computers and point-of-sales systems are used to capture and review data from stores on an hourly basis to spot new trends as early as possible. This data includes sales and inventory data and anecdotal information gleaned by sales assistants as they chat with customers and as the sales assistants gather unsold items that customers tried on, but left in fitting rooms. All this data is sent to Zara's headquarters where it is carefully analyzed by design teams who decide what new designs will be prototyped and produced in small quantities to see what sells.

In addition, inventory optimization models help the company determine the quantities and sizes of existing items that should be delivered to each store. Zara's outstanding supply chain (which includes information systems as an integral component) has led to improved customer satisfaction, decreased risks of overstocking the wrong items, reduced total costs, and increased sales.

Coles is the second largest supermarket chain in Australia. The company employs advanced analytics to improve consumer demand forecasts, and it uses sophisticated customer-loyalty analysis tools to deepen its understanding of customer buying patterns so it can plan effective marketing programs.

Coles uses a supplier portal that supports effective working relationships between the chain and its more than 3,000 suppliers, providing better coordination and communications, which, in turn, help the company reduce costs and achieve consistent delivery times along the supply chain. The portal also provides suppliers with information they can use to assess their performance and identify opportunities of improvement.

Coles works closely with suppliers to take effective measures toward continuous improvement, including frank discussions about opportunities for improvement. All these actions have gone a long way toward reducing the number 1 complaint from Coles' customers—item stockouts. They have also paved the way for improved inventory management and better supplier relationships.

Virtual Teams and Collaborative Work

A virtual team is a group of individuals whose members are distributed geographically, but who collaborate and complete work through the use of information systems. The virtual team may be composed of individuals from a single organization or from multiple organizations. The team can consist of just two people up to hundreds of team members. One benefit of virtual teams is that they enable organizations to enlist the best people in different geographical regions to solve important organizational problems. Another benefit is that they provide the ability to staff a team with people who have a range of experience and knowledge that stems from a variety of professional experiences and cultural backgrounds.

Often, it is difficult for members of a large virtual organization to meet at a time that is convenient for everyone on the team due to the time zone differences in their various geographic locations. Thus, members of a virtual team may seldom meet face to face. However, quick communication exchanges among members can be key to project success. Thus, virtual team members may need to continually monitor their email, instant messages, and team Web site and be prepared to participate in an audio or video teleconference on short notice. See **Figure 2.4**. Virtual team members must be prepared to do work anywhere, anytime. As a result, members of a virtual team may feel that their work day never ends.



FIGURE 2.4

Group videoconference

Communications are greatly improved when participants can see one another and pick up facial expressions and body language. Thus, even with sophisticated information system tools, virtual teams still benefit from occasional face-to-face meetings. This is particularly true at the beginning of new projects when the team is just forming and defining goals, roles, and expectations on how its members will work together. Virtual organization members must also be sensitive to the different cultures and practices of the various team members to avoid misunderstandings that can destroy team chemistry. It helps if virtual team members take the time to get to know one another by sharing experiences and personal background information.

International advertising, marketing, and public relations firm Ogilvy & Mather maintains offices in 161 cities all over the world and employs some 15,000 workers. The firm promotes the brands of its multinational clients by combining local market know-how with a worldwide network of resources to build powerful campaigns that address local market needs while also reinforcing global brand identity. The firm relies on multiple virtual teams across the organization to implement this strategy.

Change in the Organization

Your organization's current products, services, and ways of accomplishing work are doomed to obsolescence. Fail to change and your competition will take away your customers and your profits. Positive change is a key ingredient for any successful organization. This section will discuss important topics related to change, including innovation, reengineering, continuous improvement, outsourcing, offshoring, and downsizing.

Innovation

Innovation is the application of new ideas to the products, processes, and activities of a firm, leading to increased value. Innovation is the catalyst for the growth and success of any organization. It can build and sustain profits, create new challenges for the competition, and provide added value for customers. Innovation and change are absolutely required in today's highly competitive global environment; without both, the organization is at risk of losing its competitiveness and becoming obsolete. The following is a list of just a few of today's most innovative products:

- Tile is an innovative new product that helps solve a universal problem that we all encounter—occasionally misplacing everyday items and wasting time trying to find them. Tile is a smartphone app combined with small devices (tiles) that consumers can stick on their keys, TV remote controls, purses, and wallets. A proximity sensor plays a musical sound through the smartphone app when you come within 100 feet of the tile, so you can walk around to see if the missing item is hiding nearby.

- Healthcare technology company iHealth has introduced several different sensors that can measure and report on a wide array of biometric data, including steps taken, distance covered, and calories burned; sleep efficiency; blood pressure; glucose level; and blood oxygen saturation level and pulse rate.
- Butterfleye offers a new, economical home security product that employs a megapixel camera smart enough to recognize you, members of your family, and even your pets. If a stranger is caught inside your home within view of the camera, Butterfleye uses your home Wi-Fi system to alert you via an app.
- NeuroMetrix created Quell, an FDA-approved device that stimulates the brain to block pain receptors for patients with chronic conditions. The device is worn around the calf and calibrated to the user's body to ensure that it delivers the exact amount of relief needed. Quell performs functions similar to existing devices that today must be surgically implanted at much higher cost.

Various authors and researchers have identified different ways of classifying innovation. A simple classification developed by Clayton Christensen, a leading researcher in this field, is to think of two types of innovation—sustaining and disruptive.

Sustaining innovation results in enhancements to existing products, services, and ways of operating. Such innovations are important because they enable an organization to continually increase profits, lower costs, and gain market share. Procter and Gamble has invested hundreds of millions of dollars into making sustaining innovations to its leading laundry detergent, Tide, which was first introduced in 1946.

The innovations include: the reformulation of Tide so it works as well in cold water as it does in hot water; the creation of concentrated Tide, which reduces packaging and distribution costs; and the addition of scent, which makes clothes smell fresher. These innovations have allowed Tide to remain one of the leading detergents for several decades, with around \$5 billion in worldwide annual sales.⁷ The brand currently holds a 38 percent share of the North American laundry soap business.

A disruptive innovation is one that initially provides a lower level of performance than the marketplace has grown to accept. Over time, however, the disruptive innovation is improved to provide new performance characteristics, becoming more attractive to users in a new market. As it continues to improve and begins to provide a higher level of performance, it eventually displaces the former product or way of doing things.

The cell phone is a good example of a disruptive innovation. The first commercial handheld cell phone was invented in 1973. It weighed 2.5 pounds, had a battery life of less than 30 minutes, cost more than \$3,000, and had extremely poor sound quality.

Compare that with today's ubiquitous cell phones that have one-tenth the weight, one-fifteenth the cost, and 25 times longer battery life; smartphones can not only place calls but also serve as a camera, a video recorder, and a handheld computer that can run applications and access the Internet.

Reengineering and Continuous Improvement

To stay competitive, organizations must occasionally make fundamental changes in the way they do business. In other words, they must innovate and change the activities, tasks, or processes they use to achieve their goals.

Reengineering, also called **process redesign** and **business process reengineering (BPR)**, involves the radical redesign of business processes, organizational structures, information systems, and values of the organization to achieve a breakthrough in business results. See **Figure 2.5**. Successful reengineering can reduce delivery time, increase product and service quality, enhance customer satisfaction, and increase revenues and profitability.

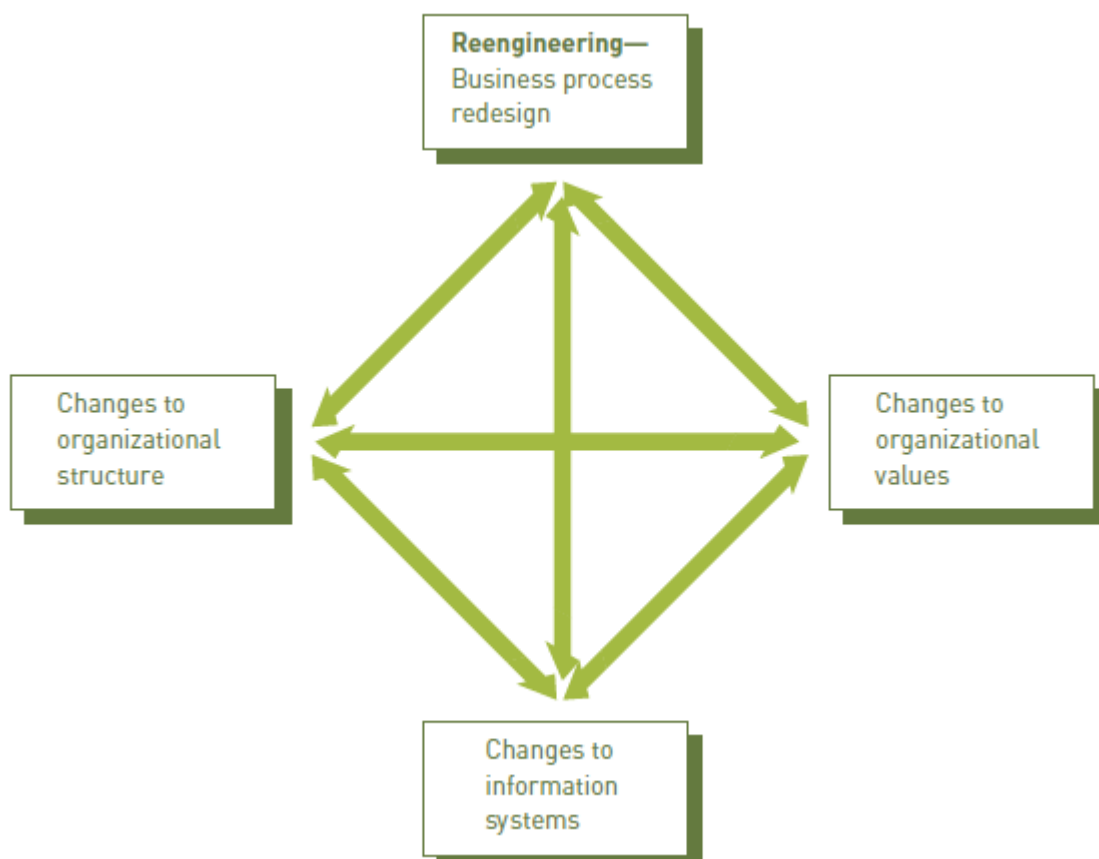


FIGURE 2.5

Reengineering

General Electric employs a strategy it calls GE Advantage to reengineer business processes to increase its speed to market, improve the quality of its products and services, reduce costs, and achieve competitive advantage. This strategy has been applied to its new product introduction process to reduce the time it takes to bring a new product to market (in some cases by more than 50 percent), improve the quality of its designs, and lower development costs.

In contrast to reengineering, the idea of continuous improvement (often referred to by the Japanese word “Kaizen”) is a form of innovation that constantly seeks ways to improve business processes and add value to products and services. This continual change will increase customer satisfaction and loyalty and ensure long-term profitability. Manufacturing companies make continual product changes and improvements. Service organizations regularly find ways to provide faster and more effective assistance to customers. By doing so, organizations increase customer loyalty, minimize the chance of customer dissatisfaction, and diminish the opportunity for competitive inroads.

Boeing has a long tradition and culture supportive of continuous improvement, which has yielded many positive results. For example, the time to assemble its popular long-range, wide-body, twin-engine 777 jet airliner has been reduced almost in half through the implementation of many small improvements—from 71 days in 1998 to just 37 days today. **Table 2.1** compares the strategies of business process reengineering and continuous improvement.

TABLE 2.1
Comparing business process reengineering
With continuous improvement



Business Process Reengineering	Continuous Improvement
Strong action taken to solve serious problem	Routine action taken to make minor improvements
Top-down change driven by senior executives	Bottom-up change driven by workers
Broad in scope; cuts across departments	Narrow in scope; focuses on tasks in a given area
Goal is to achieve a major breakthrough	Goal is continuous, gradual improvements
Often led by resources from outside the company	Usually led by workers close to the business
Information systems are integral to the solution	Information systems provide data to guide the improvement team

Outsourcing, Offshoring, and Downsizing

A significant portion of expenses for most organizations goes toward hiring, training, and compensating employees. Naturally, organizations try to control costs by determining the number of employees they need to maintain high-quality goods and services without being overstaffed. Strategies to contain these personnel costs include outsourcing, offshoring, and downsizing.

Outsourcing is a long-term business arrangement in which a company contracts for services with an outside organization that has expertise in providing a specific function. Organizations often outsource a process so they can focus more closely on their core business—and target their limited resources to meet strategic goals. Typically, the outsourcing firm has expertise and other resources that enable it to perform the service better, faster, and/or more cheaply. As a result, many companies now outsource jobs such as call center services, payroll activities, information system operations, computer support services, and security services.

Offshore outsourcing (also called **offshoring**) is an outsourcing arrangement in which the organization providing the service is located in a country different from the firm obtaining the services. Offshoring of tasks that require significant customer interaction has led to problems due to culture and language differences for some companies. As a result, many companies are reevaluating their decision to offshore their call center and customer support services, as well as the outsourcing of other activities.

Deutsche Bank has extended an offshore outsourcing agreement with Accenture to provide its procurement and accounts payable services. The original contract was signed in 2004 and was set to expire in 2016. Through its agreement with Accenture, Deutsche Bank was looking to reduce operational costs, improve spending control, and simplify its sourcing and procurement services. The arrangement

has worked well, and the two organizations recently signed a new contract to extend the initial agreement through the end of 2021.

Citizens Bank of Rhode Island signed a five-year agreement with IBM to outsource its information systems services, with the goal of increasing efficiencies, lowering costs, and improving service. The bank's information systems employees are training replacements who work for IBM in India, where the average salary of IBM workers is about \$17,000. It is estimated that some 250 to 350 current employees and existing contractors will lose their jobs at Citizen Bank as a result of the offshore outsourcing move.

Companies considering outsourcing need to take into account many factors. A growing number of organizations are finding that outsourcing does not necessarily lead to reduced costs. One of the primary reasons for cost increases is poorly written contracts that allow the service provider to tack on unexpected charges. Other potential drawbacks of outsourcing include loss of control and flexibility, the potential for data breaches of information stored on the service provider's computer hardware, overlooked opportunities to strengthen core competencies of the firm's own employees, and low employee morale. In addition, organizations often find that it takes years of ongoing effort and a large up-front investment to develop a good working relationship with an outsourcing firm. Finding a reputable outsourcing partner can be especially difficult for a small or midsized firm that lacks experience in identifying and vetting contractors.

Outsourcing part or all of a business process introduces significant risks that the service provider will introduce quality problems into the supply chain. For example, Boeing made a strategic decision to partially outsource development of the 787 Dreamliner aircraft, with a goal of reducing costs by \$4 billion and cutting development time by two years.

However, the development effort spiraled out of control—the project was ultimately billions of dollars over budget and three years behind schedule. And outsourcing complications led to severe quality challenges, including problems with the aircraft's lithium-ion batteries, which resulted in a worldwide temporary grounding of the aircraft a little more than a year after its launch.

Downsizing, a term frequently associated with outsourcing, involves reducing the number of employees to cut costs. The euphemistic term “rightsizing” is sometimes also used. When downsizing, companies usually look to downsize across the entire company, rather than picking a specific business process to downsize. Downsizing clearly reduces total payroll costs, although the quality of products and services and employee morale can suffer. Shortly after Heinz merged with Kraft in March 2015, the food and beverage company announced that it would downsize its 46,000 employees in the United States and Canada by 2,500 people to save \$1.5 billion in annual costs.

Organizational Culture and Change

Culture is a set of major understandings and assumptions shared by a group, such as within an ethnic group or a country. **Organizational culture** consists of the major understandings and assumptions for an organization. The understandings, which can include common beliefs, values, and approaches to decision making, are often not stated or documented as goals or formal policies. For example, salaried employees might be expected to check their email and instant messages around the clock and be highly responsive to all such messages.

Mark Twain said, “It’s not the progress I mind; it’s the change I don’t like.” **Organizational change** deals with how organizations successfully plan for, implement, and handle change. Change can be caused by internal factors, such as those initiated by employees at all levels, or by external factors, such as those wrought by competitors, stockholders, federal and state laws, community regulations, natural disasters, and general economic conditions.

Implementing change, such as a new information system introduces conflict, confusion, and disruption. People must stop doing things the way they are accustomed to and begin doing them differently. Successful implementation of change only happens when people accept the need for change and believe that the change will improve their productivity and enable them to better meet their customers’ needs.

The so-called **soft side of implementing change** involves work designed to help employees embrace a new information system and way of working. This effort represents the biggest challenge to successful change implementation; yet, it is often overlooked or downplayed, resulting in project failure. Indeed, both the Standish Group and Gartner, two highly respected organizations that track project implementations globally, believe that a significant contributor to project failures is overlooking the need to address employee adoption and resistance jointly.

Another resource claims that 30 to 70 percent of large information systems projects fail, at least in part, due to a failure to prepare the business users for the actual change to come.

The California Department of Consumer Affairs is made up of more than 40 entities (including multiple boards, bureaus, committees, and one commission) that regulate and license professional and vocational occupations that serve the people of California. Each year, the department processes over 350,000 applications for professional licensure along with some 1.2 million license renewals. The BreEZe project was initiated in 2009 to streamline the way the department does its business and interacts with its license applicants and consumers.

The resulting information system was intended to eliminate many paper-based processes and speed up the entire licensing process. Unfortunately, the project team failed to adequately involve the business users in the definition of the system requirements and instead made many erroneous decisions about how the system should work. The initial cost estimate for the system was \$28 million; however, as of early 2015, project costs exceeded \$37 million and less than half the licensing and regulatory boards were using the system. It is estimated that it will cost a total of \$96 million to complete the project. Much of the delay and overspending could have been avoided had the project team work better with the business users to understand their needs.

The dynamics of how change is implemented can be viewed in terms of a change management model. A **change management model** describes the phases an individual or organization goes through in making a change and provides principles for successful implementation of change. A number of models for dealing with the soft side of implementing change will now be introduced.

Lewin's Change Model

Kurt Lewin and Edgar Schein proposed a three-stage approach for change called Lewin's change model. See Figure 2.6. The first stage, unfreezing, is ceasing old habits and creating a climate that is receptive to change. Moving, the second stage, involves learning new work methods, behaviors, and systems. The final stage, refreezing, involves reinforcing changes to make the new process second nature, accepted, and part of the job.

Unfreezing Preparing for change	Moving Making the change	Refreezing Institutionalizing
Key Tasks	Key Tasks	Key Tasks
Communicate what, why, when, who, how	Motivate individuals involved or affected	Monitor progress against success criteria
Draw on others, and seek input, ideas	Coach, train, lead, encourage, manage	Establish processes, systems to institutionalize change
Define objectives, success criteria, resources, schedule, budget	Provide appropriate resources	Establish controls to ensure change is occurring
Finalize work plans	Provide on-going feedback	Recognize and reward individuals for exhibiting new behavior
Assign leaders and implementation teams		Provide feedback, motivation, additional training to individuals not exhibiting new behavior

FIGURE 2.6

Lewin's change model

Lewin's Force Field Analysis

A frequently encountered stumbling block to the successful implementation of change, including the implementation of a new system, is negative user reaction. People affected by the change may fear that their positions may be eliminated or their work altered in a way they do not like. Or they may see the introduction of a new information system as a threat to their power and influence. Such fears can lead to resentment, lack of cooperation, or outright resistance. Any of these reactions can doom a change project, no matter how carefully the rest of the project is planned.

Lewin extended his change model theory to include **force field analysis**, which identifies both the driving (positive) and restraining (negative) forces that influence whether change can occur. The **driving forces** are beliefs, expectations, and cultural norms that tend to encourage a change and give it momentum. **Restraining forces** are those that make it difficult to accept a change or to work to implement a change. For a change to occur, the strength of the driving forces must exceed the strength of the restraining forces.

This can be done in one of two ways: first, by creating new driving forces or making existing driving forces stronger and, second, by eliminating or weakening existing restraining forces.

Figure 2.7 is an example of a force-field analysis of a group of workers after they first learn that a new information system is to be installed. The feelings shown on the left side are restraining forces against the change. The feelings shown on the right side are driving forces that support the change. The length of the arrow represents the relative strength of that feeling. In this example, fear of loss of job is the strongest restraining force.



FIGURE 2.7

**Lewin's force field analysis
before addressing concerns**

Negative feelings must be reduced or eliminated in order for a new system to gain acceptance. The fear of losing one's job can be eliminated by making it clear that the individual will remain employed by the company. The fear of major changes in one's job can be reduced by allowing the individual to participate in developing one's own new job description. If this is not possible, the person should be thoroughly informed of the new job requirements and provided any necessary training. The positive impact of the change should be stressed.

It is not enough to reduce or neutralize negative feelings. Positive feelings must also be created to truly motivate the individual. Managers must take the time to explain the many tangible and intangible benefits for the organization as well as for the individual. For instance, in many information system efforts, the new system may lead to job enrichment by enabling the individual to take on more responsibility or to work in a new and more interesting way.

Figure 2.8 is an example of a force field analysis of workers after managers have effectively prepared them to accept the system. At this point, the workers should recognize several things: (1) their role is essential to the success of the system, and they are making an important contribution to the organization, (2) the development of new skills and knowledge will enhance their career growth, and (3) each individual has an important responsibility to perform within the project to secure the potential benefits for both the individual and the organization.



FIGURE 2.8

**Lewin's force field analysis
after addressing concerns**

Leavitt's Diamond

Leavitt's diamond is another organizational change model that is extremely helpful in successfully implementing change. Leavitt's diamond proposes that every organizational system is made up of four main components—people, tasks, structure, and technology—that all interact; any change in one of these elements will necessitate a change in the other three elements. Thus, to successfully implement a new information system, appropriate changes must be made to the people, structure, and tasks affected by the new system. See **Figure 2.9**.

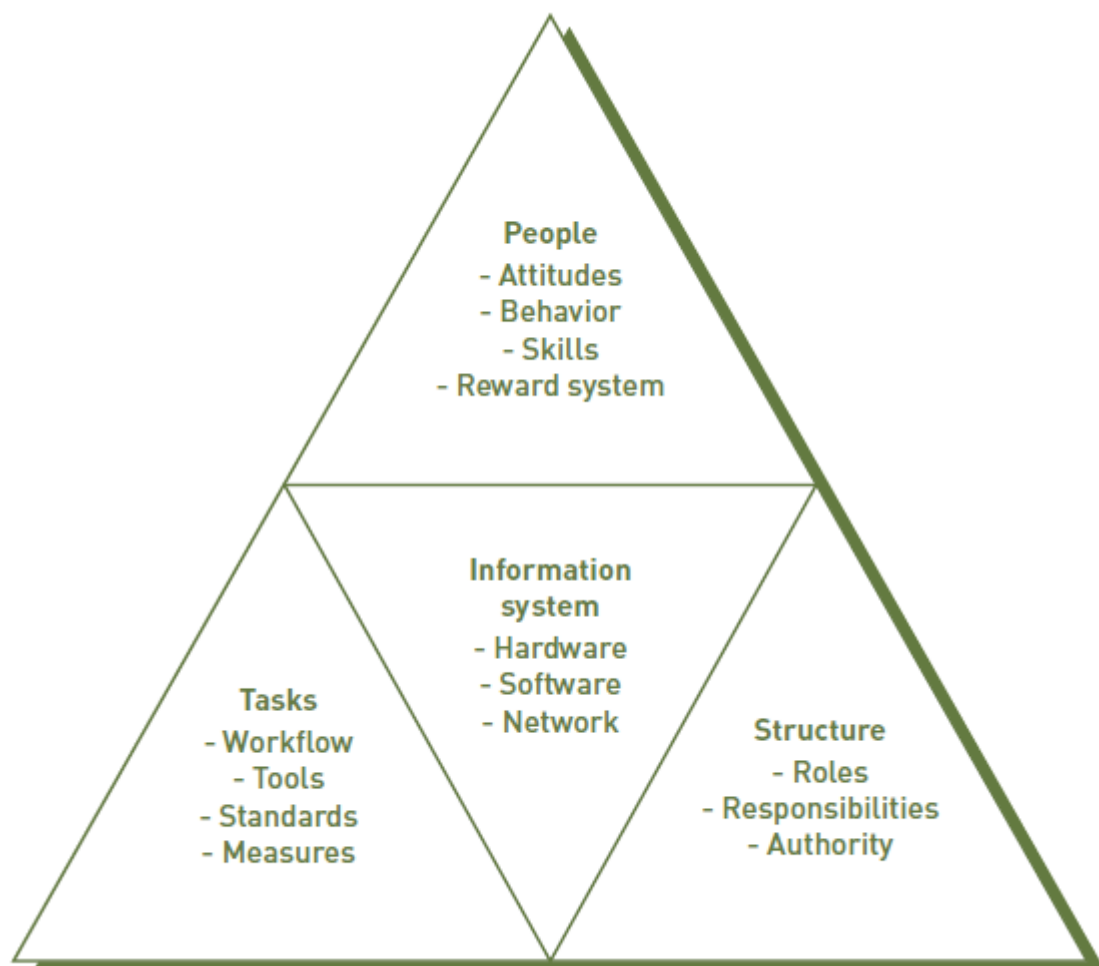


FIGURE 2.9

Leavitt's diamond

People are the key to the successful implementation of any change. They must be convinced to adopt a positive attitude about the change and be willing to exhibit new behaviors consistent with the change. This is likely to require a change in the reward system to recognize those who exhibit the desired new behaviors. Thorough training in any required new skills is also needed.

The organization structure must be modified with appropriate changes in roles, responsibilities, and lines of authority. Along with these changes are required changes in communication patterns, relationships, and coordination among those affected by the change.

The tasks required to complete the work and the flow of work between tasks also need to be changed. For each task, standards on how the work is to be performed and measures for the quality of completion need to be established. New tools may be required to perform the tasks.

As a result, the major challenges to successful implementation of an information system are often more behavioral issues than technical. Successful introduction of an information system into an organization requires a mix of both good organizational change skills and technical skills. Strong, effective leadership is required to overcome the behavioral resistance to change and achieve a smooth and successful system introduction.

Organizational learning is closely related to organizational change. All organizations adapt to new conditions or alter their practices over time—some better than others. Collectively, these adaptations and adjustments based on experience and ideas are called **organizational learning**. Hourly workers, support staff, managers, and executives learn better ways of fulfilling their role and then incorporate them into their day-to-day activities. In some cases, the adjustments can require a radical redesign of business processes (reengineering). In other cases, adjustments can be more incremental (continuous improvement). Both adjustments reflect an organization's strategy, the long-term plan of action for achieving its goals.

User Satisfaction and Technology Acceptance

Reengineering and continuous improvement efforts (including implementation of new information systems) must be adopted and used to achieve the defined business objectives by targeted users. The **technology acceptance model (TAM)** specifies the factors that can lead to better attitudes about the use of a new information system, along with its higher acceptance and usage.

See Figure 2.10. In this model, “perceived usefulness” is defined as the degree to which individuals believe that use of the system will improve their performance. The “perceived ease of use” is the degree to which individuals believe that the system will be easy to learn and use. Both the perceived usefulness and ease of use can be strongly influenced by the expressed opinions of others who have used the system and the degree to which the organization supports use of the system (e.g., providing incentives and offering training and coaching from key users). Perceived usefulness and ease of use in turn influence an individual’s attitude toward the system, which affect their behavioral intention to use the system.

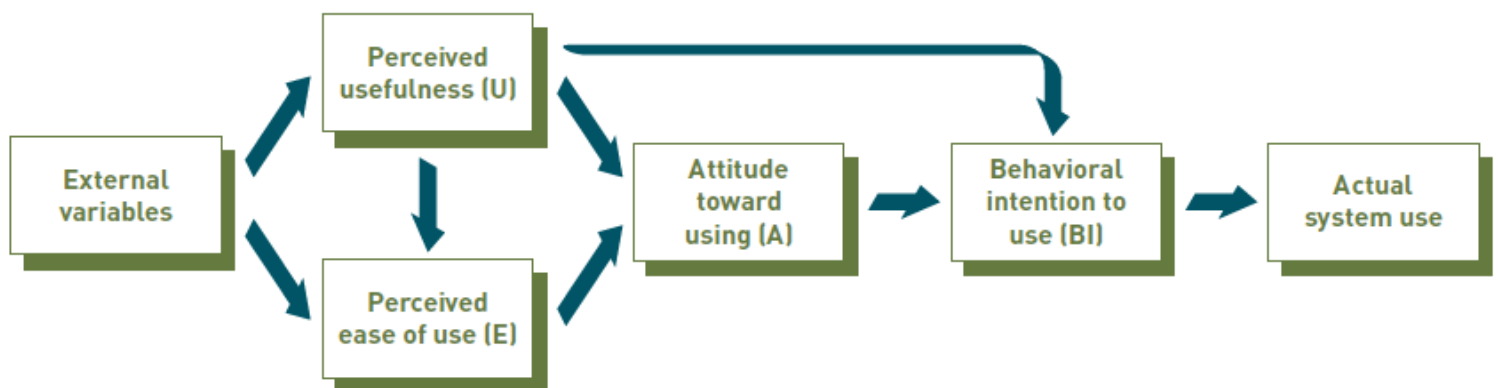


FIGURE 2.10

Technology acceptance model

Avon Products is an international manufacturer and direct seller of beauty, household, and personal care products. Avon products are sold through six million independent and mostly part-time sales representatives worldwide who sell direct to family, friends, and personal contacts. In 2013, Avon piloted a new sales system in Canada. The system was intended to streamline the ordering process through the use of iPads, which would allow the sales rep to display products to customers and then check inventory and place orders online. It was estimated that the project would generate some \$40 million per year in cost savings and increased sales.

Unfortunately, the system was poorly designed and did not meet the sales rep's expectations in terms of ease of use. Sales reps were often unable to log in to the system, and when they did get logged in, the system frequently would not accept orders, save orders correctly, or reserve inventory based on the orders placed. The system was neither useful nor easy to use.

As a result, one Avon executive sales manager estimates that as many as 16,000 Canadian sales reps quit in large part out of frustration with the new system. The pilot was such a disaster that Avon wrote off the project at a cost of nearly \$125 million.

Diffusion of Innovation Theory

The **diffusion of innovation theory** was developed by E.M. Rogers to explain how a new idea or product gains acceptance and diffuses (or spreads) through a specific population or subset of an organization. A key point of this theory is that adoption of any innovation does not happen all at once for all members of the targeted population; rather, it is a drawn-out process, with some people quicker to adopt the innovation than others. See Figure 2.11.

Rogers defined five categories of adopters, shown in Table 2.2, each with different attitudes toward innovation. When promoting an innovation to a target population, it is important to understand the characteristics of the target population that will help or hinder adoption of the innovation and then to apply the appropriate strategy. This theory can be useful in planning the roll-out of a new information system.

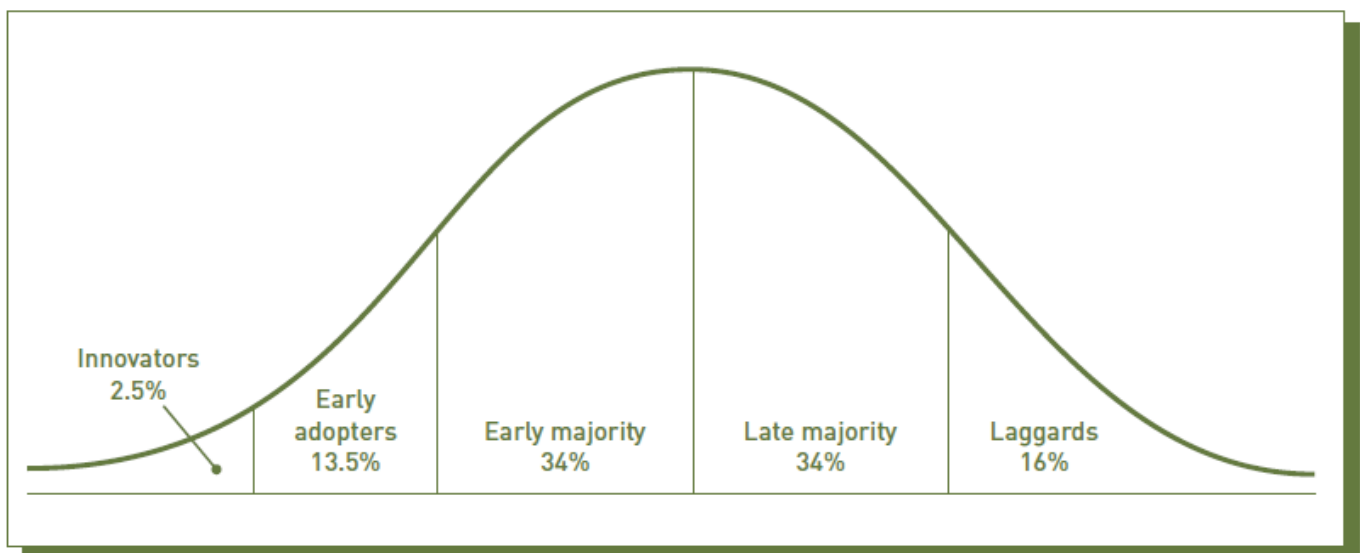


FIGURE 2.11

Innovation diffusion

TABLE 2.2
Five categories of innovation adopters

Adopter Category	Characteristics	Strategy to Use
Innovator	Risk takers; always the first to try new products and ideas	Simply provide them with access to the new system and get out of their way
Early adopter	Opinion leaders whom others listen to and follow; aware of the need for change	Provide them assistance getting started
Early majority	Listen to and follow the opinion leaders	Provide them with evidence of the system's effectiveness and success stories
Late majority	Skeptical of change and new ideas	Provide them data on how many others have tried this and have used it successfully
Laggards	Very conservative and highly skeptical of change	Have their peers demonstrate how this change has helped them and bring pressure to bear from other adopters

Prof. Dr.